

Selecting the Optimal Open-Source License for a W3C-Compliant Map Application with Monetization Goals: A Comprehensive Analysis and Recommendation

- Apache 2.0 is a permissive license enabling proprietary extensions and commercial use while keeping core code open-source.
- AGPL extends GPL copyleft to network use, requiring source code sharing for SaaS and web services, limiting proprietary monetization.
- LGPL allows proprietary software to link to open-source libraries without requiring full project open-sourcing, suitable for balancing open-source and commercial needs.
- GPL mandates all derivative works remain open-source, limiting proprietary extensions but fostering community collaboration.
- Successful projects like VLC (GPL), FFmpeg (LGPL), Red Hat, and Qt demonstrate dual licensing and open-core models to balance open-source principles with monetization.

Introduction

The challenge of selecting an open-source license for a map application built on linked open data, compliant with W3C standards, and utilizing databases such as SQLite and Oxigraph is multifaceted. The primary goal is to maintain the core project as open-source and a common good, while enabling monetization through custom services or proprietary extensions tailored to specific clients. This report provides a detailed, structured analysis of the most suitable open-source licenses—Apache 2.0, AGPL, LGPL, and GPL—comparing their implications for derivative works, commercial use, and proprietary add-ons. It also examines licensing models from projects like VLC and FFmpeg, which balance open-source cores with proprietary extensions, and offers examples of successful open-source monetization strategies. The report synthesizes legal, technical, and business perspectives to provide a clear recommendation aligned with the project's objectives.



Comparative Analysis of Open-Source Licenses

License Overview and Key Characteristics

License	Type (Copyleft/ Permissive)	Derivative Works Requirement	Commercial Use Allowed	Proprietary Extensions Allowed	Network/ SaaS Use Requirement	Compatibility with Other Licenses	Notes
Apache 2.0	Permissive	No	Yes	Yes	No	Compatible with GPLv3, not GPLv2	Includes explicit patent grant; allows proprietary modifications and commercial use
AGPL	Strong Copyleft	Yes (including network use)	Yes	No	Yes (source code must be shared)	Compatible with GPLv3	Extends GPL to SaaS; requires sharing modifications used over networks
LGPL	Weak Copyleft	Only for library modifications	Yes	Yes (for applications using library)	No	Compatible with GPLv3	Allows proprietary software to link to LGPL libraries without open-sourcing entire project
GPL	Strong Copyleft	Yes	Yes	No	No	Not compatible with Apache 2.0	Requires all derivative works to be open-source; viral copyleft effect

Detailed License Implications

Apache 2.0

Apache 2.0 is a permissive license that allows users to use, modify, distribute, and commercially exploit the software without requiring derivative works to adopt the same license. This flexibility enables proprietary modifications and commercial use, making it highly suitable for projects aiming to monetize custom services while maintaining an open-source



core. The license includes an explicit patent grant, reducing legal risks associated with patent claims, and is compatible with GPLv3 but not GPLv2 due to patent clause differences. Apache 2.0 is widely adopted in industry and open-source communities, offering a balance between freedom and commercial viability. However, its permissiveness may lead to challenges in sustaining open-source projects without additional monetization strategies such as dual licensing or paid services [1 2 3 4 5 6 7](#).

AGPL

The GNU Affero General Public License (AGPL) is a strong copyleft license that extends the GPL's requirements to software used over networks, such as SaaS applications. It mandates that any modifications to AGPL-licensed software must be shared with users interacting with the software over a network, closing the "SaaS loophole" present in GPL. This ensures that even if software is not distributed traditionally, the source code must be made available under AGPL terms. AGPL is particularly relevant in cloud computing and SaaS contexts, preventing companies from monetizing open-source software without contributing back to the community. However, AGPL's restrictive nature can hinder commercial adoption and innovation, as companies may be unwilling to share modifications. Compliance can be complex, and the license does not permit sublicensing, limiting compatibility with non-free software or permissively licensed systems [8 9 10 11 12 13 14](#).

LGPL

The GNU Lesser General Public License (LGPL) is a weak copyleft license designed as a compromise between GPL's strong copyleft and permissive licenses like MIT. It allows developers to use and integrate LGPL-licensed libraries into their own software, including proprietary software, without requiring the entire application to be open-sourced. Only modifications to the LGPL-licensed library itself must be released under LGPL terms. This makes LGPL suitable for projects where the core library should remain open-source, but the broader application may include proprietary extensions. LGPL is compatible with GPLv3, enabling code combination under GPLv3 terms, but not inherently compatible with GPLv2. LGPL fosters open-source principles while allowing commercial use and monetization through proprietary extensions or services [15 16 17 18 19](#).

GPL

The GNU General Public License (GPL) is a strong copyleft license requiring that any derivative works or software linking to GPL-licensed code must also be released under the GPL. This ensures that modifications and improvements remain open-source and accessible to the community. GPL's copyleft nature promotes collaboration and transparency but limits compatibility with proprietary software, making it incompatible with such software. The GPL's viral effect requires that any software derived from GPL code must also be licensed under GPL, which can complicate commercial adoption. GPL allows commercial use but mandates source code availability for distributed modified versions, fostering software freedom but potentially discouraging organizations seeking proprietary extensions [20 21 22 19 23](#).



Licensing Models in Successful Open-Source Projects

VLC and FFmpeg

VLC media player is licensed under GPL, ensuring that any modifications or derivative works remain open-source, fostering community collaboration. FFmpeg, a foundational library for media processing, is licensed under LGPL, allowing proprietary software to link to it without requiring the entire project to be open-source. This dual licensing approach enables FFmpeg to be used in both open-source and commercial projects, including VLC. The symbiotic relationship between VLC and FFmpeg exemplifies how open-source core libraries can support proprietary extensions and commercial use while maintaining open-source principles [24 25 26 27 28](#).

Red Hat, WordPress, Mozilla Firefox, MySQL, Qt, Metabase, MuseScore

These projects illustrate diverse monetization strategies compatible with open-source principles:

- **Red Hat:** Dual licensing model offering free Linux base and paid enterprise services.
- **WordPress:** Freemium model with free core platform and paid premium themes, plugins, and hosting.
- **Mozilla Firefox:** Combination of open-source licensing and commercial strategies, including compatible licenses like MIT and BSD.
- **MySQL:** Dual licensing under GPL and proprietary licenses, enabling commercial use alongside open-source.
- **Qt:** Dual licensing allowing open-source and proprietary development, supporting innovation and commercial integration.
- **Metabase:** Dual licensing for open-source and proprietary SaaS offerings.
- **MuseScore:** Dual licensing to support community contributions and commercial services.

These examples demonstrate that balancing open-source principles with commercial interests is achievable through dual licensing, open-core models, and offering combinations of free software and paid services [29 30 31 32 33 34 35 36](#).

Legal and Strategic Considerations

Choosing the right license involves understanding legal implications, compatibility with dependencies, and alignment with business goals. Consulting legal experts specializing in open-source licensing is crucial to navigate complexities, ensure compliance, and mitigate risks. Firms like Kirkland & Ellis, Pillsbury Law, Moorcrofts, Traverse Legal, GTC Law Group, and GalkinLaw offer expertise in open-source compliance, dual licensing, contributor agreements, and risk assessment. Their guidance can help structure a licensing strategy that balances open-source principles with commercial viability, ensuring the project's long-term success and legal soundness [37 38 39 40 41 42 43 44 45](#).



Final Recommendation

Given the project's goals—maintaining an open-source core compliant with W3C standards, utilizing open data and databases like SQLite and Oxigraph, and enabling monetization through proprietary extensions and custom services—the **Apache 2.0 license** is the most suitable choice. Its permissive nature allows proprietary modifications and commercial use while keeping the original code open-source, aligning well with the project's objectives. Apache 2.0's compatibility with GPLv3 and explicit patent grant further supports its suitability for a project that may integrate with other open-source components and requires legal clarity for commercialization.

However, to fully leverage monetization opportunities while maintaining open-source principles, consider adopting a **dual licensing model** or an **open-core model** where the core remains open-source under Apache 2.0, and proprietary extensions or services are offered under commercial licenses. This approach is exemplified by successful projects like Red Hat, Qt, and Metabase, which balance open-source and commercial interests effectively.

Engagement with legal experts specializing in open-source licensing is strongly recommended to tailor the licensing strategy to the project's specific needs, ensure compliance, and mitigate risks associated with open-source use and commercialization.

This comprehensive analysis synthesizes legal, technical, and business considerations to provide a clear, actionable recommendation aligned with the project's vision and goals.

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 - [3] [Open Source Licenses 101: Apache License 2.0 | FOSSA Blog](#)
 - [4] [Top 10 Questions About The Apache License](#)
 - [5] [Apache License 2.0](#)
 - [6] [Unveiling Apache License 2.0: A Comprehensive Summary, Exploration and Review](#)
 - [7] [Top Open Source Licenses Explained](#)
 - [8] [What Is the AGPL License? Requirements, Risks & Compliance](#)
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